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FINAL MARK

**GIRRAWEE HIGH SCHOOL
MATHEMATICS EXTENSION
2019
YEAR 11 Yearly Examination
ANSWERS COVER SHEET**

Student Number: _____

QUESTION	MARK	ME11-1	ME11-2	ME11-3	ME11-4	ME11-5	ME11-6	ME11-7
1-5	/5	✓	✓					✓
6	/20	✓	✓					✓
7	/13	✓	✓					✓
8	/25		✓					✓
9	/28					✓		✓
10	/14	✓	✓			✓		✓
TOTAL								
	/105	/52	/77			/42		/105

Year 11 Mathematics Extension 1 outcomes

A student:

- ME11-1** uses algebraic and graphical concepts in the modelling and solving of problems involving functions and their inverses
- ME11-2** manipulates algebraic expressions and graphical functions to solve problems
- ME11-3** applies concepts and techniques of inverse trigonometric functions and simplifying expressions involving compound angles in the solution of problems
- ME11-4** applies understanding of the concept of a derivative in the solution of problems, including rates of change, exponential growth and decay and related rates of change
- ME11-5** uses concepts of permutations and combinations to solve problems involving counting or ordering
- ME11-6** uses appropriate technology to investigate, organise and interpret information to solve problems in a range of contexts
- ME11-7** communicates making comprehensive use of mathematical language, notation, diagrams and graphs



GIRRAWEEEN HIGH SCHOOL

MATHEMATICS EXTENSION

YEAR 11

YEARLY EXAMINATION

• **Instructions**

- Reading Time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- For Section I, use the multiple-choice answer sheet
- For questions in section II, show relevant mathematical reasoning and /or calculations
- Marks may be deducted for careless or badly arranged work

Section I (5 marks)

Answer Questions 1 – 5

Use the multiple-choice answer sheet on page 5 for questions 1 – 5

1. Which of the following is the inverse function of $f(x) = 2x^{\frac{1}{5}}$?

A. $f^{-1}(x) = \frac{x^{\frac{1}{5}}}{32}$

B. $f^{-1}(x) = \frac{x^5}{32}$

C. $f^{-1}(x) = \frac{x^5}{16}$

D. $f^{-1}(x) = \frac{x^5}{2}$

2. The polynomial $P(x) = x^3 - 4x^2 - 6x + k$ has a factor $x - 2$.
What is the value of k ?

A. 2

B. 12

C. 20

D. 36

3. How many arrangements of the word *OLYMPIC* are possible if the *C* and the *L* are to be together in any order?

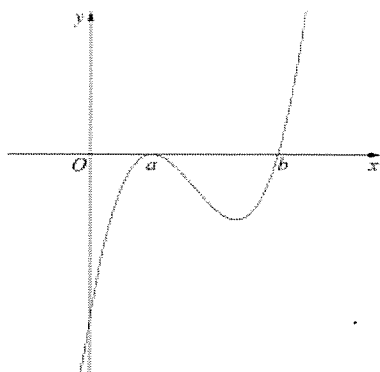
A. $5!$

B. $6!$

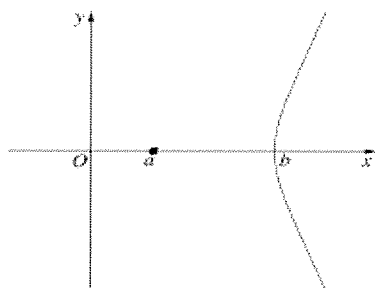
C. $2 \times 5!$

D. $2 \times 6!$

4. The diagram shows the graph of the function $y = f(x)$.



A second graph is obtained from the graph of $y = f(x)$.



Which equation best represents the second graph?

- A. $y^2 = |f(x)|$
 - B. $y^2 = f(x)$
 - C. $y = \sqrt{f(x)}$
 - D. $y = f(\sqrt{x})$
5. A curve is represented by the following parametric equations $x = t + \frac{1}{t}$, $y = t^2 + \frac{1}{t^2}$.

The Cartesian equation of the curve is:

- A. $y^2 = x - 2$
- B. $y^2 - x^2 = 2$
- C. $y = x^2 - 2$
- D. $y = x - 2$

Select the alternative A, B, C or D that best answers the question.

1.	A	☐	B	☐	C	☐	D	☐
2.	A	☐	B	☐	C	☐	D	☐
3.	A	☐	B	☐	C	☐	D	☐
4.	A	☐	B	☐	C	☐	D	☐
5.	A	☐	B	☐	C	☐	D	☐

Section II (102 marks)

- Answer questions in the spaces provided.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.

Question 6 (20 marks)

a. Solve

(i) $|3x - 5| \leq 4$ [2]

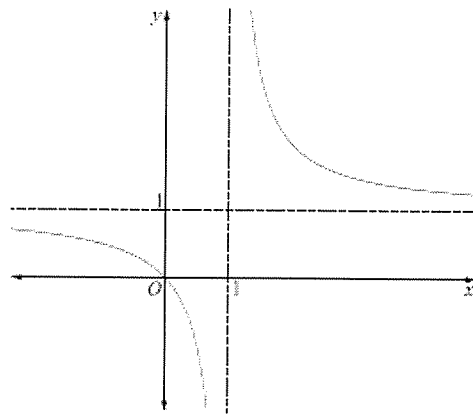
(ii) $|5x + 4| \geq 6$

[3]

$$\text{(iii)} \quad \frac{x^2 + 5}{x} > 6 \quad [3]$$

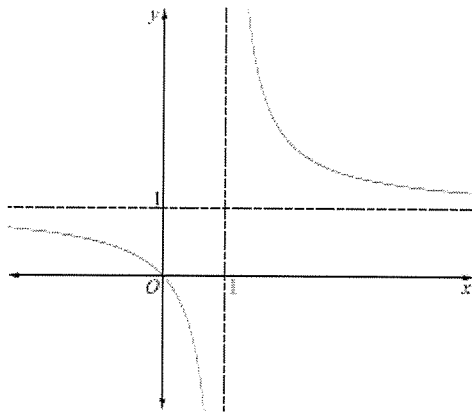
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- c. The diagram shows the graph of the function $f(x) = \frac{x}{x-1}$.

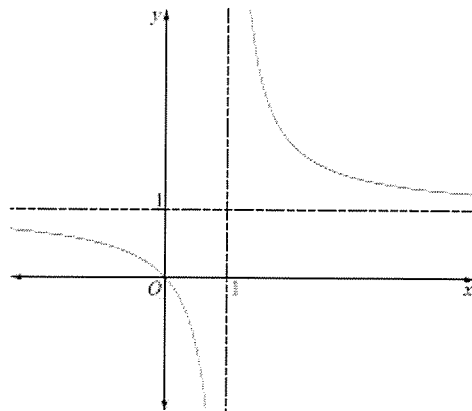


Use the graph of $y = f(x)$ to draw graphs the following transformations on the number planes provided. Clearly show all intercepts and asymptotes.

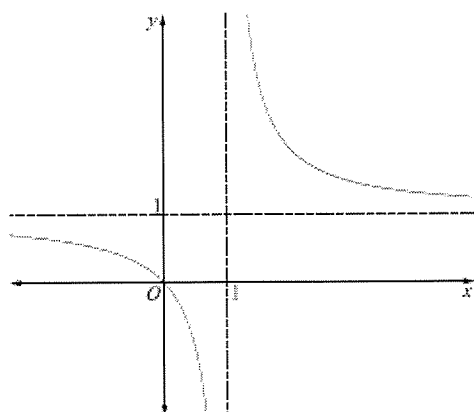
- (i) $y = |f(x)|$ [3]



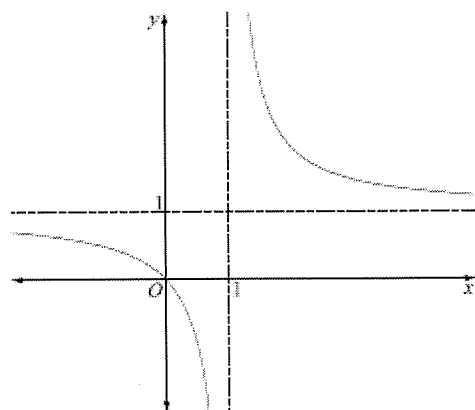
- (ii) $y = (f(x))^2$ [3]



(iii) $y = \sqrt{f(x)}$ [3]

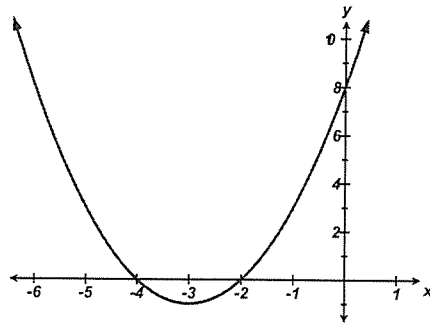


(iv) $y = f(|x|)$ [3]



Question 7 (13 marks)

- a. The graph of $f(x) = x^2 + 6x + 8$ is shown below.



- (i) Explain why the domain of $f(x)$ must be restricted if $f(x)$ is to have an inverse function.

[1]

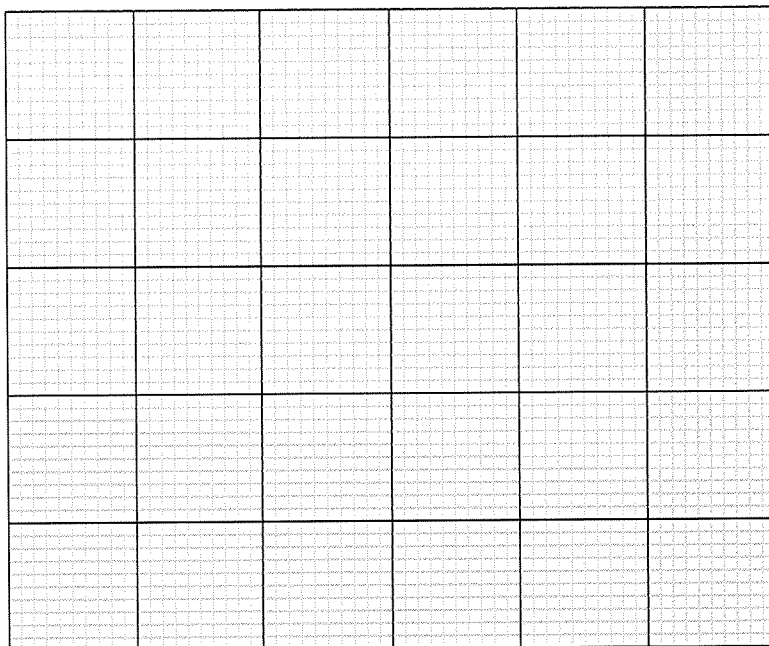
- (ii) Consider $f(x) = x^2 + 6x + 8$ for $x \geq -3$. Find the equation of $f^{-1}(x)$.

[3]

- (iii) State the domain of $f^{-1}(x)$

[1]

- (iv) Sketch the graph of $y = f(x)$ and $y = f^{-1}(x)$ for $x \geq -3$ on the grid below, clearly showing the important features. [3]



- b. The parametric equation of a graph are:

$$x = 1 - \frac{1}{t}$$

$$y = 1 + \frac{1}{t}$$

- (i) Find the Cartesian equation of the graph. [3]

(ii) Sketch the graph.

[2]



Question 8 (25 marks)

a. If α, β, γ are the roots of $2x^3 - 3x^2 + 6x + 8 = 0$, find the value of:

(i) $\alpha + \beta + \gamma$ [1]

(ii) $\alpha\beta + \beta\gamma + \alpha\gamma$ [1]

(iii) $\alpha\beta\gamma$ [1]

(iv) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ [2]

(ii) Find the other zeros.

[3]

d. Factorise $2x^3 - 5x^2 - 4x + 12 = 0$ if it has a zero of multiplicity 2 at $x = 2$.

[4]

e. Let $P(x) = (x + 1)(x - 3)Q(x) + ax + b$ where $Q(x)$ is a polynomial and a and b are real numbers. The polynomial $P(x)$ has a factor of $x - 3$.
When $P(x)$ is divided by $x + 1$, the remainder is 8.

(i) Find the values of a and b .

[3]

(ii) Find the remainder $P(x)$ is divided by $(x + 1)(x - 3)$. [1]

Question 9 (28 marks)

a. Simplify

(i) $n! + (n - 1)!$ [2]

(ii) $\frac{n!}{(n - 1)!} \cdot \frac{(n - 2)!}{(n + 1)!}$ [3]

b. Siya has five different coloured blocks. She stacks two, three, four or five blocks on top of one another to form a vertical tower.

(i) How many different towers can she form that are three blocks high? [1]

(ii) How many different towers can she form in total? [2]

c. (i) In how many different ways can the letters of the word ESTEEM be arranged? [1]

(ii) In how many ways can the letters of the word ESTEEM be arranged so that the E's are together? [2]

d. Find the number of ways in which 3 boys and 4 girls line up in a queue so that girls occupy the middle position and exactly one of the two end positions. [3]

e. (i) In how many ways can a committee of 7 be selected from a group of 18 people? [1]

(ii) In how many ways can a committee of 3 men and 4 women be selected if there are 8 men and 10 women in this group? [2]

(iii) In how many ways can a committee of 3 men and 4 women be selected from this group if particular man and a particular woman can't be in the committee together? [3]

f. Six men and six women are to be seated at a round table.

In how many ways can they be seated:

(i) if there are no restrictions? [1]

(ii) if men and women alternate? [2]

(iii) if two particular people are not to sit together?

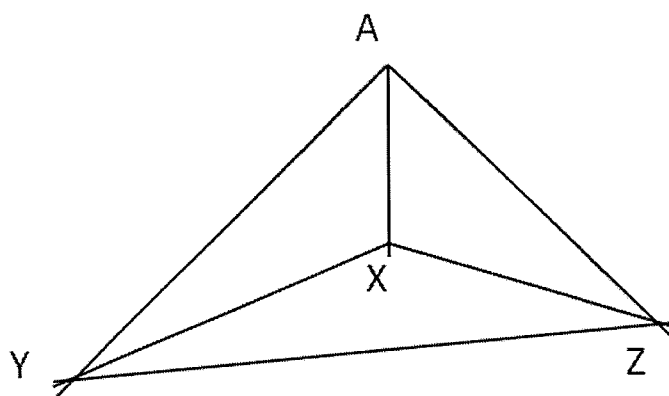
[3]

(iv) if three particular people are to sit together?

[2]

Question 10 (15 marks)

a.



The tower AX, of height h metres, is built on flat ground. X, Y and Z are points on the same horizontal plane.

From Y and Z, the angles of elevation to the top of the tower are 28° and 40° respectively. Give that $YZ = 100\text{m}$ and $\angle YXZ = 90^\circ$,

(i) show that $XY = h \cot 28^\circ$ and find a similar statement for XZ.

[2]

(ii) find the height of the tower to the nearest metre.

[3]

b. Find the inverse of $y = 3\log_e(2x + 5) - 4$.

[2]

c. Solve $x^5 + 2x^4 - 2x^3 - 8x^2 - 7x - 2 = 0$ if it has a root of multiplicity 4.

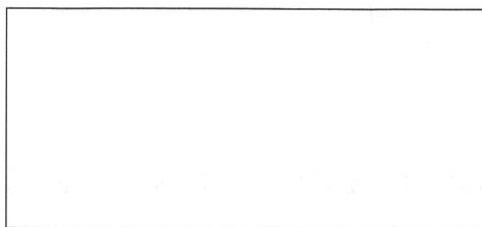
[4]

- d. A hostel has four vacant rooms. Each room can accommodate a maximum of four people. In how many ways can six people be accommodated in the four rooms? [3]

End of Examination

[illegible]

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☒ B. $f^{-1}(x) = \frac{x^5}{32}$

C. $f^{-1}(x) = \frac{x^5}{16}$

D. $f^{-1}(x) = \frac{x^5}{2}$

$$\begin{aligned} y &= 2x^{\frac{1}{5}} \\ f^{-1}(x) &: x = 2y^{\frac{1}{5}} \\ y^{\frac{1}{5}} &= \frac{x}{2} \\ y &= \frac{x^5}{32} \end{aligned}$$

2. The polynomial $P(x) = x^3 - 4x^2 - 6x + k$ has a factor $x - 2$.

What is the value of k ?

A. 2

B. 12

☒ C. 20

D. 36

$$\begin{aligned} P(2) &= 2^3 - 4(2)^2 - 6(2) + k = 0 \\ k &= 20 \end{aligned}$$

3. How many arrangements of the word *OLYMPIC* are possible if the *C* and the *L* are to be together in any order?

A. $5!$

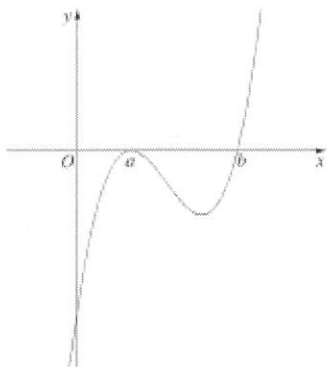
B. $6!$

C. $2 \times 5!$

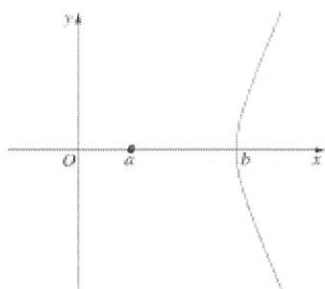
☒ D. $2 \times 6!$

$$\boxed{CL} O Y M P I$$

4. The diagram shows the graph of the function $y = f(x)$.



A second graph is obtained from the graph of $y = f(x)$.



D: $x = a, x \geq b \Rightarrow$ not A, D
 $y = \pm \sqrt{f(x)}$
 $\therefore y^2 = f(x)$

Which equation best represents the second graph?

- A. $y^2 = |f(x)|$
 B. $y^2 = f(x)$
 C. $y = \sqrt{f(x)}$
 D. $y = f(\sqrt{x})$
5. A curve is represented by the following parametric equations $x = t + \frac{1}{t}$, $y = t^2 + \frac{1}{t^2}$.
 The Cartesian Equation of the curve is:

- A. $y^2 = x - 2$
 B. $y^2 - x^2 = 2$
 C. $y = x^2 - 2$
 D. $y = x - 2$

$x^2 = \left(t + \frac{1}{t}\right)^2$
 $= t^2 + 2 + \frac{1}{t^2}$
 $\therefore x^2 - 2 = t^2 + \frac{1}{t^2}$
 $= y$

Select the alternative A, B, C or D that best answers the question.

1.	A	☐	B	☒	C	☐	D	☐
2.	A	☐	B	☐	C	☒	D	☐
3.	A	☐	B	☐	C	☐	D	☒
4.	A	☐	B	☒	C	☐	D	☐
5.	A	☐	B	☐	C	☒	D	☐

Section II (100 marks)

- Answer questions in the spaces provided.
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Question 6 (20 marks)

a. Solve

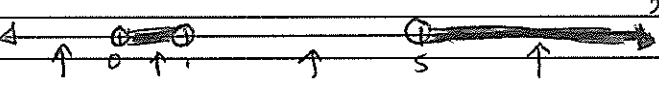
(i) $|3x - 5| \leq 4$ [2]

$$\begin{aligned} -4 &\leq 3x - 5 \leq 4 \\ 1 &\leq 3x \leq 9 \\ \frac{1}{3} &\leq x \leq 3 \end{aligned}$$

(ii) $|5x + 4| \geq 6$ [3]

$$\begin{aligned} 5x + 4 &\leq -6 & \text{or} & & 5x + 4 &\geq 6 \\ 5x &\leq -10 & & & 5x &\geq 2 \\ x &\leq -2 & & & x &\geq \frac{2}{5} \end{aligned}$$

(iii) $\frac{x^2 + 5}{x} > 6$ [3]

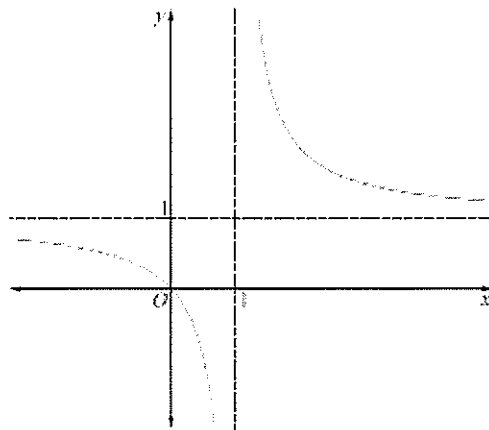
$$\begin{aligned} x \neq 0 \quad \text{so solve} \quad \frac{x^2 + 5}{x} &= 6 \\ x^2 + 5 &= 6x \\ x^2 - 6x + 5 &= 0 \\ (x - 1)(x - 5) &= 0 \\ x &= 1, 5 \end{aligned}$$


Test

$x = -1$	$x = \frac{1}{2}$	$x = 2$	$x = 6$
$\frac{(-1)^2 + 5}{-1} > 6$	$\frac{(\frac{1}{2})^2 + 5}{\frac{1}{2}} > 6$	$\frac{4 + 5}{2} > 6$	$\frac{36 + 5}{6} > 6$
$\times$	$\checkmark$	$\times$	$\checkmark$

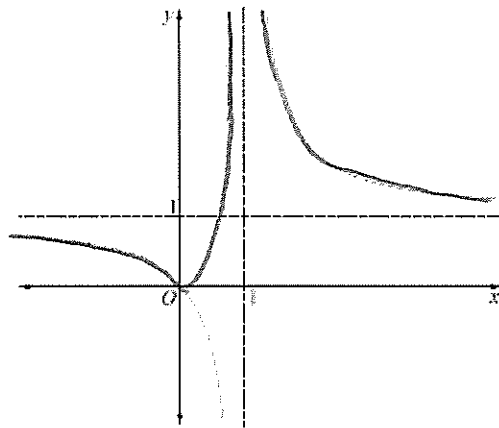
$$\therefore 0 < x < 1, x > 5$$

- c. The diagram shows the graph of the function $f(x) = \frac{x}{x-1}$.

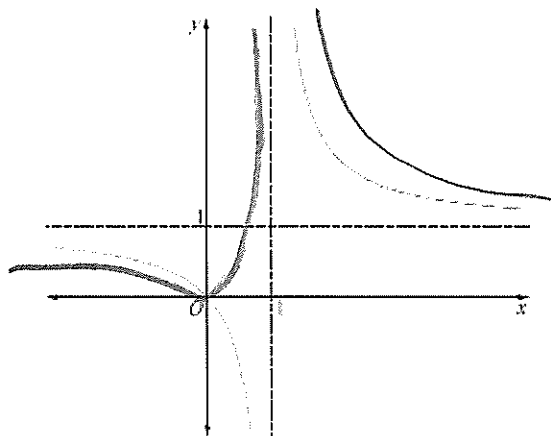


Use the graph of $y = f(x)$ to draw graphs the following transformations on the number planes provided. Clearly show all intercepts and asymptotes.

- (i) $y = |f(x)|$ [3]

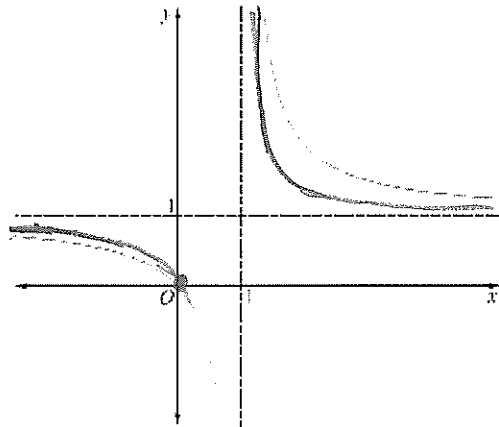


- (ii) $y = [f(x)]^2$ [3]



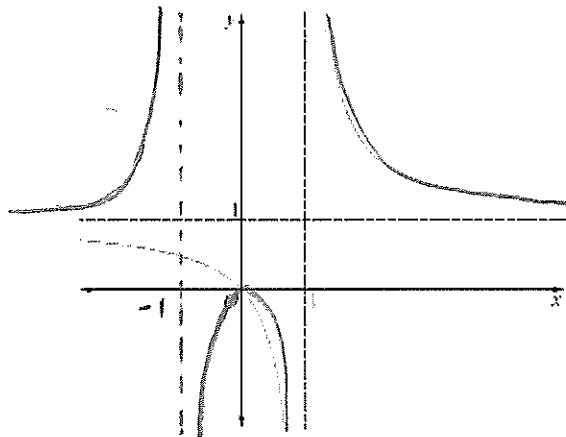
(iii) $y = \sqrt{f(x)}$

[3]



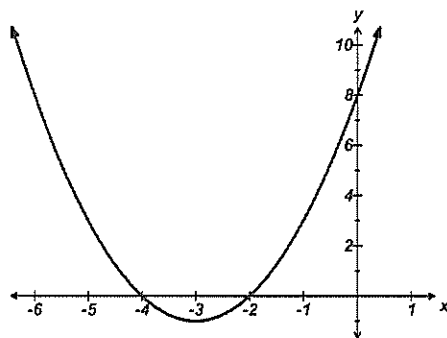
(iv) $y = f(|x|)$

[3]



Question 7 (13 marks)

- a. The graph of $f(x) = x^2 + 6x + 8$ is shown below.



- (i) Explain why the domain of $f(x)$ must be restricted if $f(x)$ is to have an inverse function. [1]

Horizontal line cuts this graph in more than 1 place
(or) The function is not a one-to-one function.

- (ii) Consider $f(x) = x^2 + 6x + 8$ for $x \geq -3$. Find the equation of $f^{-1}(x)$. [3]

$$f(x) : y = x^2 + 6x + 8$$

$$f^{-1}(x) : x = y^2 + 6y + 8$$

$$x = (y+3)^2 - 1$$

$$x+1 = (y+3)^2$$

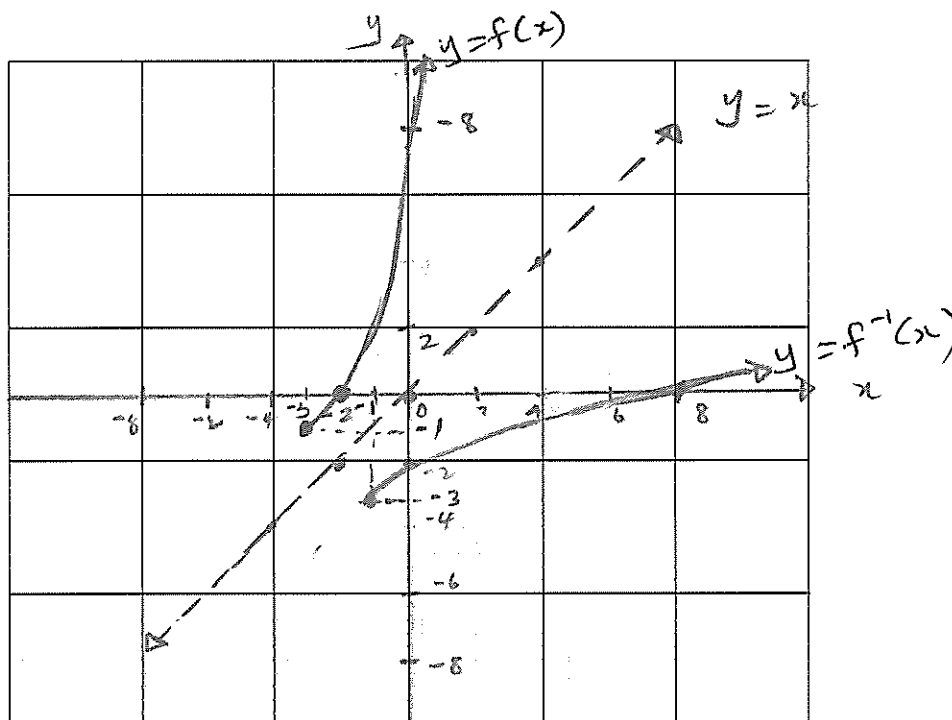
$$y+3 = \pm \sqrt{x+1}$$

$$\therefore y = -3 + \sqrt{x+1}, \text{ since } R_{f^{-1}(x)} : y \geq -3$$

- (iii) State the domain of $f^{-1}(x)$

$$D_{f^{-1}(x)} = R_{f(x)} = x \geq -1$$

- (iv) Sketch the graph of $y = f(x)$ and $y = f^{-1}(x)$ for $x \geq -3$ on the grid below, clearly showing the important features. [3]



- b. The parametric equation of a graph are:

$$x = 1 - \frac{1}{t}$$

$$y = 1 + \frac{1}{t}$$

- (i) Find the Cartesian equation of the graph. [3]

$$x = 1 - \frac{1}{t} \quad \text{--- (1)} \quad ; \quad y = 1 + \frac{1}{t} \quad \text{--- (2)}$$

From (1) $tx = t - 1$

Substitute $t = \frac{1}{1-x}$ into (2)

$$t - tx = 1$$

$$t(1-x) = 1$$

$$y = 1 + \frac{1}{\frac{1}{1-x}}$$

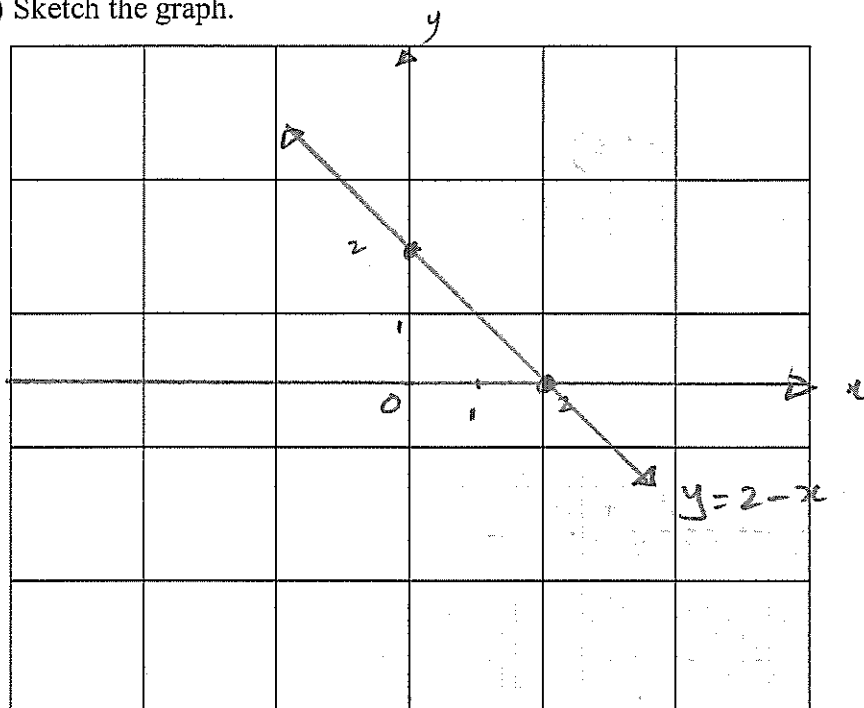
$$t = \frac{1}{1-x}$$

$$y = 1 + 1 - x$$

$$y = 2 - x$$

(ii) Sketch the graph.

[2]



Question 8 (26 marks)

a. If α, β, γ are the roots of $2x^3 - 3x^2 + 6x + 8 = 0$, find the value of:

(i) $\alpha + \beta + \gamma$ [1]

$$= -\frac{b}{a} = \frac{3}{2}$$

(ii) $\alpha\beta + \beta\gamma + \alpha\gamma$ [1]

$$= \frac{c}{a} = \frac{6}{2} = 3$$

(iii) $\alpha\beta\gamma$ [1]

$$= -\frac{d}{a} = -\frac{8}{2} = -4$$

(iv) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ [2]

$$\frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} = \frac{3}{-4} = -\frac{3}{4}$$

(v) $\alpha^2 + \beta^2 + \gamma^2$ [3]

$$= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma)$$

$$= \left(\frac{3}{2}\right)^2 - 2(3)$$

$$= \frac{9}{4} - 6 = -\frac{15}{4}$$

(vi) $\frac{\alpha}{\beta\gamma} + \frac{\beta}{\alpha\gamma} + \frac{\gamma}{\alpha\beta}$ [2]

$$= \frac{\alpha^2 + \beta^2 + \gamma^2}{\alpha\beta\gamma}$$

$$= \frac{-15/4}{-4} = \frac{15}{16}$$

- c. $x^3 + 3x^2 + \alpha x - 1 = 0$ has one root equal to the reciprocal of another.
Find the roots. [3]

Roots : $\alpha, \beta, \frac{1}{\beta}$

product of roots : $\alpha = 1$

$\therefore$ roots = $1, \beta, \frac{1}{\beta}$

Sum of roots : $1 + \beta + \frac{1}{\beta} = -3$

$$\beta + \beta^2 + 1 = -3\beta$$

$$\beta^2 + 4\beta + 1 = 0$$

$$\beta = \frac{-4 \pm \sqrt{12}}{2} = \frac{-4 \pm 2\sqrt{3}}{2}$$

$$= -2 \pm \sqrt{3}$$

$\therefore$ Roots : $1, \sqrt{3}-2, \frac{1}{\sqrt{3}-2}$

or $1, -\sqrt{3}-2, \frac{1}{-\sqrt{3}-2}$

- b. Consider the polynomial $P(x) = x^3 - 2x^2 - 5x + 6$

- (i) Show that $x = 1$ is a zero of $P(x)$. [1]

$$P(1) = 1 - 2 - 5 + 6 = 0$$

$\therefore x = 1$ is a zero

(ii) Find the other zeros.

[3]

$$\begin{array}{r}
 x^2 - x - 6 \\
 x-1 \overline{) x^3 - 2x^2 - 5x + 6} \\
 \underline{-(x^3 - x^2)} \\
 -x^2 - 5x \\
 \underline{-(-x^2 + x)} \\
 -6x + 6 \\
 \underline{-(-6x + 6)} \\
 0
 \end{array}$$

$$\begin{aligned}
 \therefore P(x) &= (x-1)(x^2 - x - 6) \\
 &= (x-1)(x-3)(x+2)
 \end{aligned}$$

Other zeros : -2, 3

c. Factorise $2x^3 - 5x^2 - 4x + 12 = 0$ if it has a zero of multiplicity 2 at $x = 2$.

[4]

Roots : 2, 2, α

$(x-2)^2$ is a factor

$$\text{Sum of roots : } 2 + 2 + \alpha = \frac{5}{2}$$

$$\alpha = -\frac{3}{2}$$

$\therefore (2x+3)$ is a factor

$$\therefore 2x^3 - 5x^2 - 4x + 12 = (x-2)^2(2x+3)$$

d. Let $P(x) = (x+1)(x-3)Q(x) + ax + b$ where $Q(x)$ is a polynomial and a and b are real numbers. The polynomial $P(x)$ has a factor of $x-3$. When $P(x)$ is divided by $x+1$, the remainder is 8.

(i) Find the values of a and b .

[3]

$(x-3)$ is a factor $\therefore P(3) = 0$

$$\therefore 3a + b = 0 \quad \text{--- (1)}$$

When $\div$ by $(x+1)$, remainder = 8 $\therefore P(-1) = 8$

$$\therefore -a + b = 8 \quad \text{--- (2)}$$

$$\textcircled{1} - \textcircled{2} \quad \therefore 4a = -8$$

$$a = -2$$

Substitute $a = -2$ into $\textcircled{1}$

$$-6 + b = 0$$

$$b = 6$$

$$\therefore a = -2, b = 6$$

(ii) Find the remainder $P(x)$ is divided by $(x+1)(x-3)$.

[1]

$$P(x) = (x+1)(x-3)Q(x) + (-2x+6) \text{ (from (i))}$$

$$\therefore \text{Remainder when divided by } (x+1)(x-3) \\ = -2x+6$$

Question 9 (28 marks)

a. Simplify

(i) $n! + (n-1)!$ [2]

$$= n(n-1)! + (n-1)! \\ = (n-1)!(n+1)$$

(ii) $\frac{n!}{(n-1)!} \cdot \frac{(n-2)!}{(n+1)!}$ [3]

$$= \frac{n(n-1)!(n-2)!}{(n-1)!(n+1)n(n-1)(n-2)!} \\ = \frac{1}{(n+1)(n-1)}$$

b. Siya has five different coloured blocks. She stacks two, three, four or five blocks on top of one another to form a vertical tower.

(i) How many different towers can she form that are three blocks high? [1]

$$5 \times 4 \times 3 \text{ (or)} {}^5P_3 = 60 \\ = 60$$

(ii) How many different towers can she form in total? [2]

$${}^5P_2 + {}^5P_3 + {}^5P_4 + {}^5P_5 \\ = 20 + 60 + 120 + 120 \\ = 320$$

c. (i) In how many different ways can the letters of the word ESTEEM be arranged? [1]

$$\frac{6!}{3!} = 120 \quad \text{6 letters, 3 Es}$$

- (ii) In how many ways can the letters of the word ESTEEM be arranged so that the E's are together? [2]

$\boxed{EEE} _ _ _$

$$4! = 24$$

- d. Find the number of ways in which 3 boys and 4 girls line up in a queue so that girls occupy the middle position and exactly one of the two end positions. [3]

$G \dots G \dots B \sim B \dots G \dots G$

$$2 \times {}^4C_2 \times {}^3C_1 \times 4! \times 2! = 1728$$

- e. (i) In how many ways can a committee of 7 be selected from a group of 18 people? [1]

$${}^{18}C_7 = 31824$$

- (ii) In how many ways can a committee of 3 men and 4 women be selected if there are 8 men and 10 women in this group? [2]

$${}^8C_3 \times {}^{10}C_4 = 11760$$

- (iii) In how many ways can a committee of 3 men and 4 women be selected from this group if particular man and a particular woman can't be in the committee together? [3]

Total ways — ways both chosen

$$11760 - {}^7C_2 \times {}^9C_3 = 9996$$

- e. Six men and six women are to be seated at a round table.

In how many ways can they be seated:

- (i) if there are no restrictions? [1]

$$11! = 39916800$$

- (ii) if men and women alternate? [2]

$$5! \times 6!$$

$$= 86400$$

(iii) if two particular people are not to sit together? [3]

11 groups

$$2 \text{ sit together} = 10! \times 2! = 7257600$$

$$\therefore \text{Not sit together} = 11! - 10! \times 2!$$

$$= 10! (11 - 2) = 9 \times 10! = 32659200$$

(iv) if three particular people are to sit together? [2]

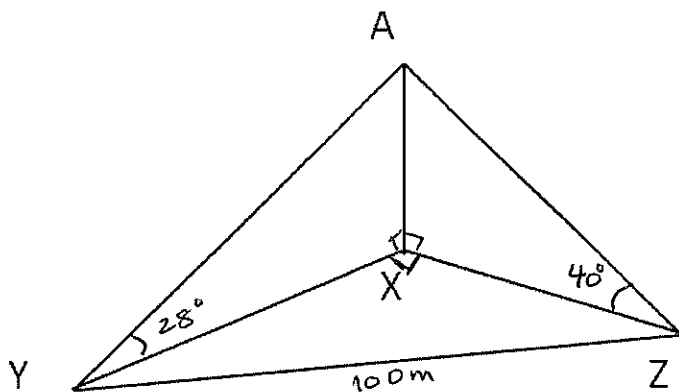
10 groups

$$9! \times 3!$$

$$= 2177280$$

Question 10 (15 marks)

a.



The tower AX, of height h metres, is built on flat ground. X, Y and Z are points on the same horizontal plane.

From Y and Z, the angles of elevation to the top of the tower are 28° and 40° respectively. Give that $YZ = 100\text{m}$ and $\angle YXZ = 90^\circ$,

(i) show that $XY = h \cot 28^\circ$ and find a similar statement for XZ. [2]

In $\triangle AX Y$

$$\tan 28^\circ = \frac{h}{XY}$$

$$XY = \frac{h}{\tan 28^\circ}$$

$$XY = h \cot 28^\circ$$

Similarly, using $\triangle AX Z$

$$XZ = h \cot 40^\circ$$

(ii) find the height of the tower to the nearest metre.

[3]

Using Pythagoras' theorem in $\triangle XYZ$

$$YZ^2 = XY^2 + XZ^2$$

$$(100)^2 = (h \cot 28^\circ)^2 + (h \cot 40^\circ)^2$$

$$= h^2 \cot^2 28^\circ + h^2 \cot^2 40^\circ$$

$$(100)^2 = h^2 (\cot^2 28^\circ + \cot^2 40^\circ)$$

$$h = \frac{100}{\sqrt{\cot^2 28^\circ + \cot^2 40^\circ}}$$

$$= 44.913 \dots$$

height of tower = 45 metres.

b. Find the inverse of $y = 3 \log_e(2x + 5) - 4 = f(x)$

[2]

$$f^{-1}(x) : x = 3 \log_e(2y + 5) - 4$$

$$\frac{x+4}{3} = \log_e(2y+5)$$

$$e^{\frac{x+4}{3}} = 2y+5 \Rightarrow y = \frac{1}{2} \left(e^{\frac{x+4}{3}} - 5 \right)$$

c. Solve $x^5 + 2x^4 - 2x^3 - 8x^2 - 7x - 2 = 0$ if it has a root of multiplicity 4.

[4]

$$P(x) = x^5 + 2x^4 - 2x^3 - 8x^2 - 7x - 2 \quad (\text{root of multiplicity 4})$$

$$P'(x) = 5x^4 + 8x^3 - 6x^2 - 16x - 7 \quad (\text{same root of multiplicity 3})$$

$$P''(x) = 20x^3 + 24x^2 - 12x - 16 \quad (\text{--- --- } x^2)$$

$$P'''(x) = 60x^2 + 48x - 12 \quad (\text{--- --- --- } x^1)$$

$$\text{Roots of } 60x^2 + 48x - 12 = 0$$

$$5x^2 + 4x - 1 = 0$$

$$(5x-1)(x+1) = 0$$

$$x = \frac{1}{5}, -1$$

$$P(-1) = (-1)^5 + 2(-1)^4 - 2(-1)^3 - 8(-1)^2 - 7(-1) - 2$$

$$= 0$$

$\therefore x = -1$ is the zero of multiplicity 4

$\therefore (x+1)^4$ is a factor

$$\therefore P(x) = (x+1)^4(x-2) \quad \text{By observation}$$

$$\therefore x = -1, -1, -1, -1, 2$$

- d. A hostel has four vacant rooms. Each room can accommodate a maximum of four people. In how many ways can six people be accommodated in the four rooms? [3]

$$\text{Total number of ways (no maximum)} = 4^6 = 4096$$

$$\text{Number of ways with max 4} = \text{Total} - (\text{ways to have 5} + \text{ways to have 6})$$

$$\text{No. of ways to have 5 in one room} = {}^6C_5 \times {}^4C_1 \times {}^3C_1 = 72$$

$$\text{No. of ways to have 6 in one room} = 4$$

$$\therefore \text{No. of ways with 4 maximum in a room.}$$

$$= 4096 - (4 + 72)$$

$$= \underline{4020}$$

End of Examination